

SHAUNAK RANE

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SUMMARY

Passionate AI and tech enthusiast with hands-on experience in software development and data science. Skilled in Python, machine learning, mathematics, and statistics, with a strong foundation in areas such as linear algebra, probability, and calculus. Adept at applying mathematical and statistical concepts to solve complex data problems and build intelligent systems. Always eager to learn, innovate, and contribute to AI-driven projects.

WORK EXPERIENCE

Univitt Ai Technologies

April 2025 - July 2025

Software Project Trainee

Developed a Python-based backend using FastAPI to track inventory and automatically send alerts for items nearing expiration, reducing potential spoilage by an estimated 25%. Implemented an SQL database to manage user data, inventory, and donation logistics, ensuring efficient data retrieval and management. Built and integrated a machine learning model using Random Forest Regression to predict inventory demand and optimize stock utilization.

PROJECTS

Designed and implemented a Food Management System using a full-stack architecture. Developed a responsive frontend using HTML, CSS, and JavaScript for data input, visualization, and user interaction. Built backend services using Python (Flask) to handle CRUD operations, RESTful APIs, and business logic. Integrated a relational database (MySQL/SQLite) for structured storage of food inventory, users, and transactions. Implemented authentication, input validation, and role-based access control. Optimized database queries and ensured secure data flow between frontend and backend. The project highlights proficiency in full-stack development, API design, and database management.

Github - <https://github.com/Shاونakrane914/Food-Management>

Developed an agentic AI system for automated misinformation detection using NLP. Implemented multiple autonomous agents for text ingestion, claim extraction, semantic similarity analysis, and fact verification. Used transformer-based embeddings to compare claims against trusted sources and generate confidence scores. Designed an agent orchestration pipeline enabling inter-agent communication and decision flow. Exposed the system through REST APIs for real-time inference. The project demonstrates skills in NLP, multi-agent architectures, and scalable AI system design.

Github - <https://github.com/Shاونakrane914/Misinformation>

Developed a machine learning-based crop yield prediction system to help farmers and organizations forecast agricultural output more accurately. Implemented multiple models and achieved the best performance using Random Forest Regression, leveraging Python, Pandas, Scikit-learn, and Matplotlib for model development and visualization. The project demonstrated strong integration of mathematics, statistics, and AI concepts to create a practical, data-driven solution for improving agricultural planning and resource management.

Github- <https://github.com/Shاونakrane914/Yield-Ai>

SKILLS

- Technical Skills: Python, FastAPI, Machine Learning, Data Science, SQL, Software Development, Mathematics, Statistics, Problem Solving
- Certifications: — Generative Ai, Learning C Programming,Ai for everyone, Introduction to LLM,etc